Price Prediction using Multi-Modal Embeddings

Introduction

This project aims to predict product prices using multi-modal embeddings derived from images and text descriptions. By leveraging embeddings from **SigLIP2-Giant** for images and **Mistral-7B-v0.1** for text embeddings, we train a robust regression model to predict prices. The final model combines these embeddings using a custom neural network architecture incorporating Encoders and Wide Deep architecture.

Machine Learning Approach

Data Preprocessing

- **Image Embeddings:** Inferred using the SigLIP2-Giant model. Images were resized to 384x384 for consistency.
- **Text Embeddings:** Generated using the Mistral 7B model. Text descriptions were tokenized and embedded using mean pooling.
- **Feature Combination:** Image and text embeddings were concatenated to form a unified feature matrix.

Model Architecture

WideDeepTwoEncoderMLP: A custom neural network architecture with two encoders. Utilises Linear layers with ReLU and BatchNorm. Also uses Dropout layers to improve generalisation. The Model consists of 7.5M parameters:

- **Image Encoder**: Processes image embeddings.
- **Text Encoder**: Processes text embeddings.
- **Wide Path**: Directly uses raw features for additional context.
- **Attention Mechanisms**: Applied to each encoder to weigh the resultant vectors, emphasizing important features.
- **Regression Head**: Combines all paths to predict prices.

Training Strategy

- **Cross-Validation**: 6-fold cross-validation ensures robust evaluation.
- **Loss Function**: Mean Absolute Error (MAE) was used for training.
- **Optimiser:** Adam optimiser was used.
- **Log Transformation**: Applied to the target variable to handle long-tail distributions.

Experiments

Baseline 1: Simple MLP Head

1. **Image Features**: ViT embeddings.
2. **Text Features**: BERT embeddings.
3. **Architecture**: A simple MLP head combining the embeddings for prediction.

Baseline 2: WideDeep Model

1. **Image Features**: ViT embeddings.
2. **Text Features**: MiniLM embeddings.
3. **Architecture**: A WideDeep model head, which combines wide (linear) and deep (non-linear) paths for prediction.

Baseline 3: Ensemble (5-Fold)

1. **Image Features**: ViT embeddings.
2. **Text Features**: Mistral embeddings.
3. **Architecture**: Ensemble of previous models trained using 5-fold cross-validation, leveraging the diversity of predictions for better generalization.

Baseline 4: Two-Encoder Model

1. **Image Features**: SigLIP2Giant embeddings.
2. **Text Features**: Mistral7B-v0.1 embeddings.
3. **Architecture**: Two separate encoders for image and text features, combined with a Wide-Deep prediction head. This approach allows the model to process image and text features independently in the Deep branch and jointly in the Wide branch before merging them for the final prediction. We switched to 6-Fold ensemble as well.

Conclusion

The final model, **WideDeepTwoEncoderMLP**, demonstrated significant improvements over simpler architectures by effectively combining multi-modal embeddings. The use of SigLIP2-Giant for image embeddings and Mistral 7B for text embeddings provided a strong foundation for price prediction. Attention mechanisms further enhanced the model's ability to focus on relevant features.

Key Takeaways:

- Multi-modal embeddings significantly improve price prediction performance.
- SigLIP2-Giant and Mistral 7B embeddings proved to be well-suited for this task.
- Future work could explore:
    - Fine-tuning embeddings for domain-specific tasks.
    - Integration with FAISS for fast similarity searches.

Team Members

- Romit Mohane
- Shardul Junagade
- Swayam Borate
- Umang Shikarvar